FIGURE 11: ANSYS MULTIPHYSICS SIMULATION ARCHITECTURE & VERIFICATION

Deterministic Electromagnetic Antenna (Ansys Maxwell / HFSS) & Thermal CFD (Ansys Fluent) Modeling Pipelines

ANSYS MAXWELL / HFSS 3D ELECTROMAGNETIC FEA

Antenna 3D CAD & Boundary Setup

[SIMULATED]

Geometry: 865 MHz Quarter-Wave Copper Whip (L = 82.5 mm, r = 0.5 mm)

Substrate & Casing: IP65 ABS Enclosure (epsilon_r = 2.8, tan_delta = 0.005)

Radiation Boundary: Air box padded lambda/4 with PML absorbing boundary

Mesh Refinement & Adaptive Solver

[SIMULATED]

Mesh: Adaptive tetrahedral meshing (18,420 elements converged)

Excitation: 50 Ohm Lumped Wave Port at SMA feedpoint

Frequency Sweep: 840 MHz to 890 MHz discrete sweep (0.5 MHz steps)

Convergence: Delta S < 0.01 across 4 consecutive adaptive passes

Simulated RF Output Metrics

[SIMULATED]

Return Loss: S_11 = -22.4 dB at 865.0625 MHz (Target < -15 dB) [SIMULATED]

Voltage Standing Wave Ratio: VSWR = 1.16 : 1 (Near ideal 50 Ohm match)

Radiation Pattern: Omnidirectional Doughnut in H-plane (Gain: +2.15 dBi)

Radiation Efficiency: 91.4% (minimal dielectric loss in ABS wall)

ANSYS FLUENT COMPUTATIONAL FLUID DYNAMICS (CFD)

Langstroth Hive CFD Domain

[SIMULATED]

Geometry: 10-Frame Langstroth Brood Box (465 x 375 x 240 mm)

Thermal Core: Brood cluster heat source (15W metabolic heat generation)

Materials: Pinewood walls (k = 0.12 W/m-K), Wax comb (k = 0.25 W/m-K)

Mesh & Navier-Stokes Solver

[SIMULATED]

Mesh: 1.2M Polyhedral cells with boundary-layer inflation at comb faces

Physics: Coupled Navier-Stokes + Boussinesq Buoyancy Natural Convection

Boundary Conditions: Ambient air 15.0°C, entrance vent natural draft (0.1 m/s)

Turbulence Model: k-omega SST low Reynolds number formulation

Simulated Thermal Results

[SIMULATED]

Core Temperature Retention: Stable 34.5°C at central Frame 4/5 boundary.

Thermal Stratification: 19.2°C temperature gradient from core to bottom board.

Validates Sensor Placement: TMP117 in core measures true biological status;

Peripheral DS18B20 probes measure convective boundary dissipation.

[MEASURED] [VALIDATED] [CALCULATED] [SIMULATED] [DEMONSTRATED]